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Evaluating and Improving Flight Derivation from ADS-B Data

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1. Introduction

Flight datasets can be derived from surveillance data such as ADS-B, operational records, or a combination of both. Operational records may contain errors, be unavailable for some flights, omit censored aircraft, and be less accessible than ADS-B data. Existing publicly available ADS-B-derived flight datasets include OpenSky [1] and ADS-B Exchange [2]. Comprehensive ground-truth flight datasets are not readily available. Evaluation of existing datasets has been limited to aggregate flight count comparisons [3]. We present an algorithm that outperforms existing flight-derivation methods on the reference dataset constructed here.

2. Methods

2.1 Deriving the Reference Flight Dataset

Operational records were obtained from the FAA SWIM Flight Data Publication Service (SFDPS) and the Bureau of Transportation Statistics (BTS) On-Time Performance dataset. We recorded the SFDPS feed for six months and derived flights from its messages using an algorithm, `sfdps_to_flights.py`. March 1, 2026, was selected for evaluation because ADS-B Exchange and OpenSky published ADS-B data and derived flight datasets for that date. BTS contained 17,609 flights from 4,718 aircraft. The SFDPS-derived dataset contained 27,486 flights from 12,559 aircraft. Only aircraft for which the two datasets agreed were retained. Of 4,441 aircraft present in both datasets, 3,136 had matching flight sequences comprising 9,852 flights. The matched sequences were then validated against ADS-B Exchange data using `flights_match_adsb.py`, eliminating cases in which both operational sources were incomplete or agreed on the same incorrect flight sequence, including shared omissions of flights outside FAA-managed airspace. This produced a reference dataset of 2,377 flights from 607 aircraft. ADS-B Exchange data was used because it yielded the largest number of validated flight sequences.

2.2 Flight Segmentation Algorithm

`readsb` [4] uses a finite-state machine based on temporal gaps, reported ground status, and altitude to divide an aircraft's ADS-B trace into flight legs. This approach was adapted to derive flights.

2.3 Airport Identification Using LightGBM

A LightGBM model identifies airports using the nearest defined ADS-B data points to the beginning or end of each flight segment, together with airport data from OurAirports. Each candidate airport is represented by features defined in `features.py`. All features are defined relative to each candidate airport; no absolute aircraft or airport coordinates are used. This design is intended to reduce overfitting and support generalization to airports not in the training data. The model was trained using SFDPS-derived flights validated against ADSB.lol ADS-B data beginning on June 1, 2026.

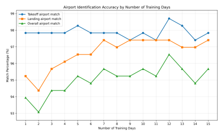

Figure 1. Airport-identification match rates for 6,250 ADS-B-validated, SFDPS-derived flights from February 1, 2026.

3. Results

For comparison, airport-identification results from a nearest-airport baseline are also provided.

Table 1. Flight-segmentation and airport-identification performance against the reference dataset for March 1, 2026. Airport-identification values include only correctly segmented flights.

ADS-B source	Segmentation F1	LightGBM takeoff	LightGBM landing	Nearest takeoff	Nearest landing	Published F1	Published takeoff	Published landing
OpenSky	0.992	97.7%	96.9%	96.0%	95.5%	0.989	84.8%	76.2%
ADS-B Exchange	1.000	100.0%	100.0%	100.0%	99.9%	0.986	100.0%	100.0%
ADSB.lol	0.990	99.4%	99.4%	99.2%	98.6%	—	—	—

Generated by: `python src/flights/evaluation/evaluation.py -test-src algorithm algorithm-no-airports opensky adsbx -adsb-src adsblol adsbx opensky -gold-src ground_truth -use_all_icaos`

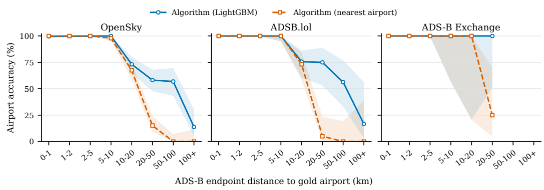

Figure 2. Airport-identification accuracy as a function of the distance between the gold airport and nearest ADS-B position in flight segment.

4. Conclusion

The segmentation algorithm and model-based airport-identification approach achieved greater ADS-B-to-flight derivation accuracy than the published datasets, and the model outperformed a nearest-airport baseline. Evaluation remains limited by the lack of comprehensive reference datasets covering more challenging flight trajectories and cases of incomplete ADS-B coverage.

Acknowledgement

I thank Martin Strohmeier for encouraging me to make this submission and for his valuable feedback.

Open data statement

All raw data except historical SFDPS messages are freely downloadable.

Reproducibility statement

All code and data, except for ADS-B records, are in https://github.com/PlaneQuery/planequery-flights-algorithm repo. ADS-B data is automatically downloaded from the respective networks.

References

[1] Martin Strohmeier, Xavier Olive, Jörn Lübbe, Matthias Schäfer, and Vincent Lenders. "Crowdsourced air traffic data from the OpenSky Network 2019–2020". In: *Earth System Science Data* 13 (2021), pp. 357–366. DOI: 10.5194/essd-13-357-2021.

[2] ADS-B Exchange. *Sample Data*. URL: https://web.archive.org/web/20260724010901/https://www.adsbexchange.com/data-products/sample-data/#:~:text=mm/dd/xx-,Flights,-Complete%20flight%20records (visited on 07/28/2026).

[3] Quinten Goens, Rainer Koelle, Enrico Spinielli, Tatjana Bolić, Andrew Cook, and Martin Strohmeier. "The Open Performance Data Initiative: A Foundation Supporting the European Open Science Alliance for ATM Research". In: *Proceedings of the 14th SESAR Innovation Days*. Rome, Italy, 2024. URL: https://www.sesarju.eu/sites/default/files/documents/sid/2024/papers/SIDs_2024_paper_100%20final.pdf.

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